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**Aragya Goyal**  
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## EDUCATION

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- University of Pittsburgh (Swanson School of Engineering/Frederick Honors Col.)** **Pittsburgh, PA**  
• *B.S. - Computer Engineering (Autonomous Systems Focus); GPA: 3.99/4.00* *August 2022 - April 2026*  
**Courses:** Robotic Control, Data Structures and Algorithms, Embedded Processors, Microelectronics

## PROFESSIONAL EXPERIENCE

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- ESTAT Actuation** **Pittsburgh, PA**  
• *Robotics Developer Co-op* *May 2025 - Present*
    - Building and commissioning multiple robotic test stands expanding clutch testing capacity by 133%, covering mechanical assembly, electrical wiring, programming, and documentation for long-term use and potential sales to external customers.
    - Directing the assembly and delivery of 6 custom battery boxes for the Suzuki Moqba robot at the Japan Mobility Show. Overseeing quality control, leading debugging efforts to resolve heating issues, and ensuring a professional final build.
    - Testing new generation rotary clutch prototypes to improve reliability and evaluate longevity. Creating documentation explaining testing procedures, and using these prototypes to inform design decisions for next generation production clutches.
  - Carnegie Mellon University Robotics Institute** **Pittsburgh, PA**  
• *Undergraduate Roboticist Researcher* *April 2023 - Present*
    - ZOË 2 Rover:** ([tinyurl.com/zoe2rover](https://tinyurl.com/zoe2rover))
      - Assisting in the development of software architecture and the robotic controller implementation for a novel 4-wheeled rover with a unique passive steering mechanism.
      - Enabling low-level software development for the rover using ROS2 and C++ to interface with motor controllers and encoders via CANOpen protocol.
    - Underwater Snake Robot:** ([tinyurl.com/humrsCMU](https://tinyurl.com/humrsCMU))
      - Implemented High-Frequency Injection methods in Brushless DC thrusters to achieve control at low/zero speeds thus reducing minimum speed by 80% allowing for reduced oscillations in station-keeping control loops.
      - Deployed station-keeping feature using AprilTags, IMU readings, and Nested PID Controllers to perform robot state-estimation underwater.
      - Conducted major repairs on the buoyancy module of the robot and assisted in continual electrical/hardware maintenance.
    - Apple's E-Waste Recycling Project:** ([tinyurl.com/applecmu](https://tinyurl.com/applecmu))
      - Labeled datasets consisting of 4000+ images for Machine Learning Models to detect screws in e-waste images.
      - Integrated ROS1 and Python packages to track AprilTags using a Realsense camera for localization of UR3 robotic arm.
      - Manufactured custom AprilTags using lasercutters and sheet metal manufacturing methods.

## LEADERSHIP & ACTIVITIES

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- Robotics and Automation Society (University Rover Challenge)** **Pittsburgh, PA**  
• *Chief Engineer/Integration Lead* *August 2022 - Present*
    - Leading a multidisciplinary team of 30 students in engineering a rover, coordinating integration efforts across mechanical, electrical, software, and bioscience teams. ([tinyurl.com/roverimages](https://tinyurl.com/roverimages))
    - Mentoring younger students in Onshape, KiCAD, and general manufacturing processes through hands-on building.
    - Securing and managing over \$7,000 in funding to support team development and future growth.
    - Proposed and spearheaded the development of a prototype robotic hand utilizing pneumatics. ([tinyurl.com/hydraarm](https://tinyurl.com/hydraarm))

## PROJECTS

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- Micromouse Robot:** Designed, assembled, and programmed a custom Micromouse robot through iterative PCB development, integrating motor control, IMU, IR sensors, and a display system with low-level embedded programming. Implemented PID control loops for navigation. Placed 3rd in competition at Stony Brook University IEEE conference. ([tinyurl.com/goyalmm](https://tinyurl.com/goyalmm))
  - 555 Timer Roulette Board:** Designed a PCB LED Roulette Wheel using 555 timer and CD4017 counter, featuring dynamic speed control using capacitive timing and a 3D-printed enclosure. ([tinyurl.com/555roulette](https://tinyurl.com/555roulette))
  - VoltVitals Embedded Board:** Designed and prototyped an ESP32 based circuit with web interface to measure home outlet voltage and report over Wi-Fi, collaborating with FirstEnergy engineers to enable early outage detection. ([tinyurl.com/voltvitals](https://tinyurl.com/voltvitals))

## PUBLICATIONS

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- M. Nye, A. Raji, A. Saba, E. Erlich, R. Exley, **A. Goyal**, A. Matros, R. Misra, M. Sivaprakasam, M. Bertogna, D. Ramanan, and S. Scherer, **"BETTY Dataset: A Multi-modal Dataset for Full-Stack Autonomy,"** in Proc. IEEE Int. Conf. Robot. Autom. (ICRA), 2025. [Online]. Available: ([arxiv.org/abs/2505.07266](https://arxiv.org/abs/2505.07266))

## SKILLS AND AWARDS

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- Software:** Linux, ROS1, ROS2, Github, Python, C++, C, MATLAB, ARM and RISC-V Assembly, CANOpen, Google Colab
  - Electrical:** Altium, KiCAD, LTSpice, Soldering, Arduino, ESP32
  - Mechanical:** Solidworks, Onshape, Milling, Lathe, MIG Welding, Laser Cutting, General Shop Tools
  - Awards:** Micromouse Competition 3rd place, SSOE Design Expo 1st place Art of Making, FSAE Innovation Award, Eagle Scout

### CMU Research

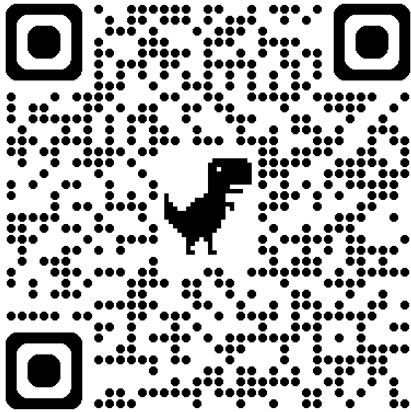


Figure 1: Zoë Rover

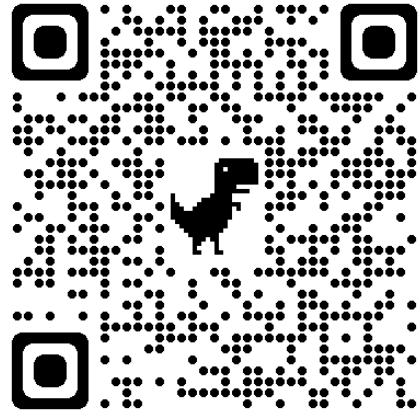


Figure 2: Underwater Snake Robot

### Finished Projects

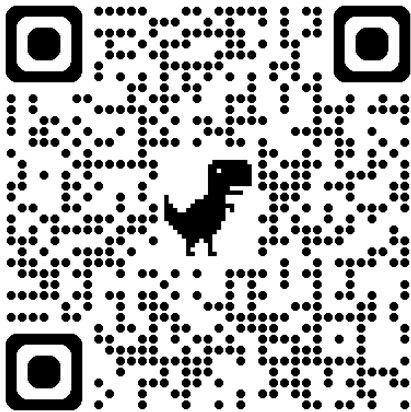


Figure 3: STM32 Elevator Simulator

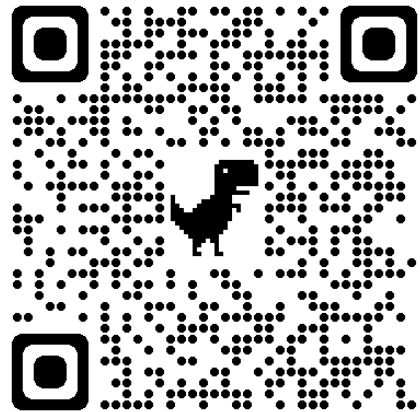


Figure 4: Custom Cane

### High School Robotics

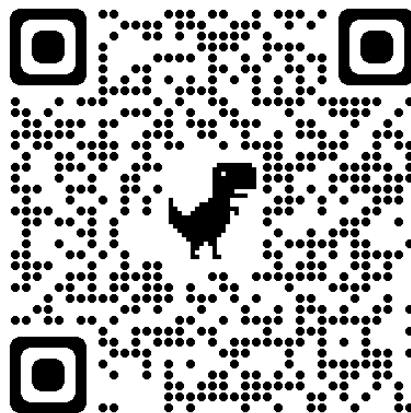


Figure 5: Dewbot XVII

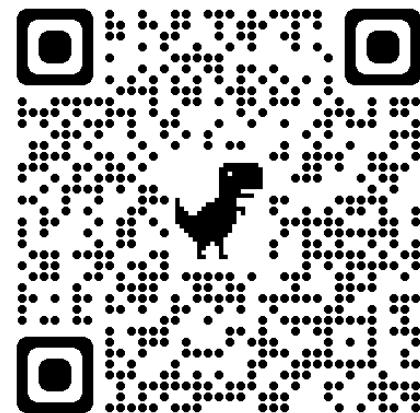


Figure 6: VEX Robotics